# 04 · Picking funds from a retail menu: does past performance help, and what do fees cost?

This report's text is generated automatically from the code's outputs. The main output is the quantitative part: data, checks, models and tables. This study was built for this portfolio from public CVM data.

## Question

An investment committee must choose which funds to offer on a menu (a company savings plan, a platform shortlist). Two questions decide the rule: do last year's winners keep winning, and do cheaper funds leave investors better off?

## Result

• Last year's best fifth of equity funds beat the worst fifth by 1.11 points the following year, with a standard error of 4.61: no usable signal.

• The slope of next-year on this-year excess return is 0.094 for equity and 0.182 for fixed income, both significant but small.

• In fixed income, 39.3% of the best quintile stays best the next year, against 20% by chance; in equity, 28.0%.

• Each extra point of expense ratio goes with 0.735 points less net excess return in equity and 0.380 in multi-asset, both significant at 1%; in fixed income the slope is 0.061 and not significant.

• The cheapest third of equity funds (median expense ratio 0.16%) returned -4.99 points over the CDI the next year, against -7.18 for the most expensive third (median 2.82%).

• Over 10 years, the median equity investor who picks from the cheapest third ends with 0.72 of what the CDI would have given, against 0.69 for a random pick and 0.66 for last year's best quintile.

• Picking from the cheapest third and skipping last year's worst quintile gives the best median in every class: 0.74 in equity, 0.88 in multi-asset and 0.96 in fixed income.

Reading: past returns barely help; fees do. A menu rule of "cheapest third within each class, minus last year's worst" beats picking at random or chasing winners, but in 2021-2025 even that trailed the CDI. Four year pairs is a short history: these are associations in one period, not a law.

## Data checks

7 pass, 5 warnings, 0 fail (details in output/data_checks.json).

# Main tables

## Next-year excess return over the CDI (p.p.) by quintile of this year's excess return (average of the four year pairs)

| | Equity | Multi-asset | Fixed income |
|---|---|---|---|
| Q1 (worst this year) | -3.34 | -0.67 | -1.85 |
| Q2 | -2.24 | -2.09 | -1.36 |
| Q3 | -3.01 | -2.23 | -0.55 |
| Q4 | -1.07 | -1.28 | -0.24 |
| Q5 (best this year) | -2.23 | -1.08 | -1.71 |
| Q5 minus Q1 | 1.11 | -0.42 | 0.14 |
| | (4.61) | (2.68) | (2.36) |
| Slope: next on this year | 0.094** | 0.094** | 0.182*** |
| | (0.040) | (0.045) | (0.065) |
| Fund-years | 3677 | 5314 | 4619 |

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## Share of funds that stay in the same quintile next year (chance = 20%)

| | Equity | Multi-asset | Fixed income |
|---|---|---|---|
| Best quintile stays best (%) | 28.0 | 29.2 | 39.3 |
| Worst quintile stays worst (%) | 31.6 | 28.0 | 33.3 |
| Worst quintile closed next year (%) | 1.1 | 1.8 | 0.9 |

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## Fees and next-year net excess return over the CDI (p.p.)

| | Equity | Multi-asset | Fixed income |
|---|---|---|---|
| Expense ratio (slope) | -0.735*** | -0.380*** | 0.061 |
| | (0.189) | (0.127) | (0.080) |
| Performance fee (dummy) | -5.120 | 1.550 | -0.050 |
| | (3.528) | (3.273) | (0.653) |
| Observations | 3432 | 4774 | 4411 |
| R^2 | 0.365 | 0.019 | 0.066 |
| Cheapest third: next-year excess | -4.99 | -1.72 | -1.30 |
| Cheapest third: median expense ratio | 0.16 | 0.07 | 0.04 |
| Middle third: next-year excess | -4.15 | -1.48 | -1.07 |
| Middle third: median expense ratio | 1.22 | 1.10 | 0.36 |
| Most expensive third: next-year excess | -7.18 | -2.59 | -1.53 |
| Most expensive third: median expense ratio | 2.82 | 2.95 | 1.17 |

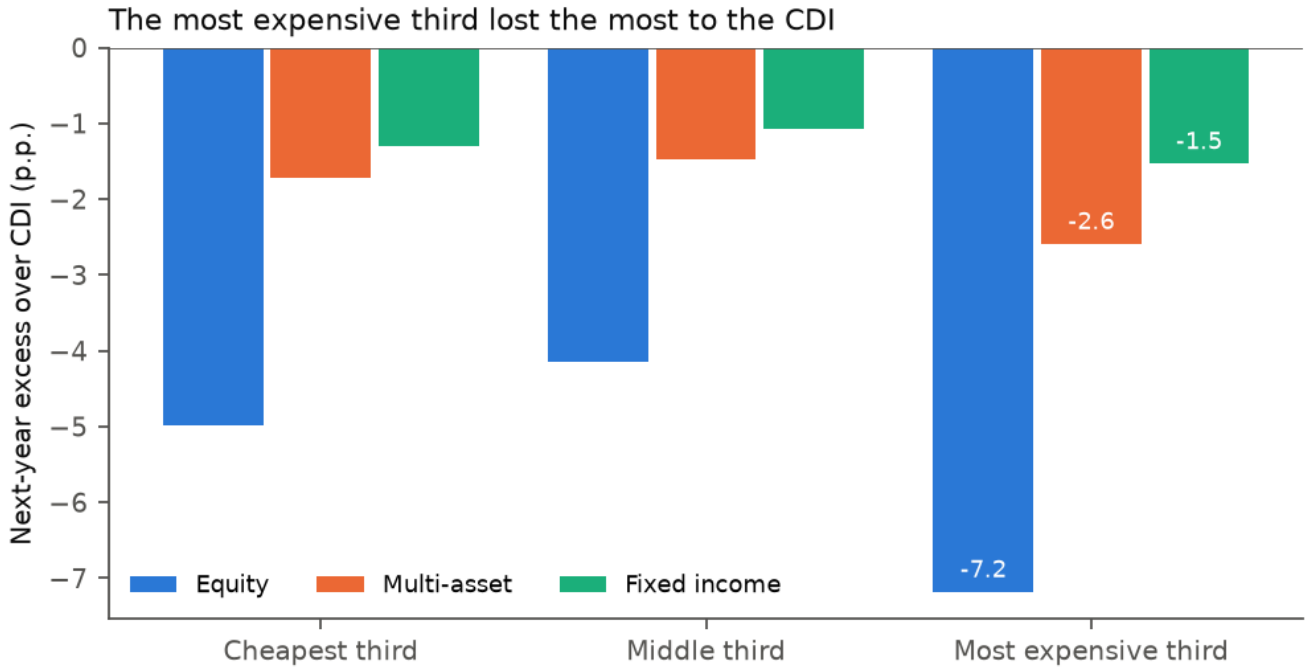
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## Monte Carlo over 10 years: wealth relative to staying in the CDI (1.00 = same as the CDI), 10000 paths

| Class: rule | Median | P5 | P95 | Beats CDI (%) |
|---|---|---|---|---|
| Equity: Random fund | 0.69 | 0.23 | 1.89 | 27 |
| Equity: Cheapest third | 0.72 | 0.24 | 2.01 | 30 |
| Equity: Last year's best quintile | 0.66 | 0.20 | 2.07 | 29 |
| Equity: Cheapest third, not in worst quintile | 0.74 | 0.26 | 1.94 | 31 |
| Multi-asset: Random fund | 0.85 | 0.47 | 1.44 | 28 |
| Multi-asset: Cheapest third | 0.87 | 0.40 | 1.50 | 32 |
| Multi-asset: Last year's best quintile | 0.82 | 0.35 | 1.81 | 32 |
| Multi-asset: Cheapest third, not in worst quintile | 0.88 | 0.54 | 1.42 | 30 |
| Fixed income: Random fund | 0.92 | 0.72 | 1.05 | 14 |
| Fixed income: Cheapest third | 0.93 | 0.64 | 1.07 | 22 |
| Fixed income: Last year's best quintile | 0.86 | 0.62 | 1.07 | 15 |
| Fixed income: Cheapest third, not in worst quintile | 0.96 | 0.75 | 1.05 | 27 |

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# Main figures




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Monte Carlo: median (dot) and 5th-95th percentile (line); 1.00 = same as the CDI




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